

QRC / ARCS: Final Plan for a Rigorous Case-Study Paper

Reconciling the document-level review with the code-level agentic audit, and a single action plan going forward

Prepared for	Akash Kishore — QRC / ARCS project (22ET49, DSCE / VTU)
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Supersedes	“Novelty, Publishability & QACT Improvement Assessment” (18 Sept 2026) — not retracted, but updated throughout by the findings below
New inputs	Three code-level audits from agentic tools run directly against the project repository (structure map, mechanistic V3/V4/QACT analysis, and an independently-executed reproducibility audit)
Directive this document follows	Reframe the paper as a rigorous case study — methodology, optimization, pros and cons — not a novel-discovery claim

1. What the Deep-Dive Changes

My original assessment was built from your project status report plus external literature — a document-level, outside-in view. The three audits you just ran actually executed code inside the repository: they re-ran evaluations, rebuilt an isolated test environment, and checked whether saved numbers reproduce. That's a fundamentally stronger form of evidence than anything in my first pass, and it changes the plan in real ways, not just in detail. Three findings matter more than anything in my original report:

- **The headline V4 number is a cherry-picked seed, not the honest result.** “V4 Seed 3, MAE 1510 ± 367 Hz” is the best of five seeds. The 5-seed aggregate — the number that should have been the headline all along — is MAE 2205 ± 518 Hz, $84.8\% \pm 7.6\%$ near-resonance. Seed 3's number is real and reproduces on rerun; the problem is that it was reported alone, as “the authoritative model,” without the aggregate alongside it. This is a bigger integrity risk than anything in my original document, and it needs fixing before anything else.
- **The QACT/Qiskit failure is a code bug, not quantum-domain difficulty.** I originally treated the sim-to-sim gap as an open research question to characterize. The audits found something more specific and more useful: the classical and Qiskit observation vectors have the same length (5) but the elements mean different things in each — Stable-Baselines3 loads the mismatched checkpoint without complaint. An isolation test that fixed only this (kept the physics different, fixed everything else) recovered most of the missing performance. This reframes the entire QACT section of the paper.
- **The saved QACT numbers do not reproduce and the pipeline is currently broken.** Rerunning the exact committed code gave materially different numbers than what's saved (2176 Hz reported vs. 2732 ± 1839 Hz on rerun; 3237 ± 4581 Hz reported vs. 3670 ± 6128 Hz on rerun), two scripts fail outright on a missing import, and the pipeline has hardcoded paths from a different machine. None of the currently saved QACT results can go into the paper as-is — not because they're wrong in spirit, but because they're not currently reproducible, which is a harder requirement.

None of this invalidates the reframing you've decided on — if anything it strengthens the case for it. A “rigorous case study” framing does not need a clean headline win; it needs honestly reported, reproducible numbers and a clear account of what was learned, including from the things that went wrong. You now have unusually good material for exactly that, provided the fixes in §3 happen first.

2. Reconciling the Two Analyses

Where the two reviews land on the same conclusion from different evidence, and where the code-level audit adds something the document-level review could not have found:

Question	My original (document + literature)	Agentic audit (code-level)	Net position
Is the project novel?	No — RL-vs-PID resonance/frequency tracking, including on real superconducting qubits with PPO specifically, is already published (Fösel 2018, Baum 2021, Ding 2023, and others).	No — gradient/derivative observation augmentation is textbook control-theory practice, and the reward already contains a gradient-shaped term, weakening even the mechanistic novelty claim.	Both hold and stack: not novel at the domain level, and not cleanly novel at the mechanism level either. Reframing as a case study sidesteps both objections.
Is “sim-to-real” the right term for QACT?	No — it's simulator-to-simulator; flagged as a labeling problem.	Implicitly confirms — frames it correctly throughout as “transfer,” and the literature it cites (Tobin 2017, Zhao 2020) is sim-to-	Confirmed. Relabel everywhere; see §10.

Question	My original (document + literature)	Agentic audit (code-level)	Net position
		real/sim-to-sim robustness literature, applied correctly.	
Why does QACT underperform?	Speculative — recommended checking normalization parity, adding noise models, fine-tuning instead of zero-shot.	Diagnosed directly: observation-vector semantic misalignment is the dominant cause (proven by an isolation experiment), with a secondary, smaller effect from a stricter near-resonance threshold caused by an arbitrary Rabi-frequency choice.	The agentic audit answers a question I could only gesture at. Use its diagnosis, not my speculation, going forward.
Is the classical V3→V4 result solid?	Not evaluated at this level of detail.	Yes, with two caveats: metrics are partly reward-aligned (lean on MAE), and the gradient-in-observation vs. gradient-in-reward confound has not been isolated.	Solid foundation, but not yet ready to be the headline claim without the confound-clearing ablation (§6) and without reporting the aggregate, not Seed 3 alone.
What's the strongest defensible result available?	Proposed: documented failure modes (A) + honest null result (B) + characterized transfer variance (C), combined.	Identifies a specific, ready-made version of the same idea: the V4a/V4b finding that naive gradient encodings score worse than no gradient at all.	Converge on the same shape of answer; the agentic audit supplies the concrete headline result my version was missing. See §7.
Role of Multisim/Simulink?	Validates the analog front-end (Stages 1/3/4) only; cannot substitute for hardware or the RL/QACT results.	Same conclusion, more specific: validate the Lorentzian shape, amplitude thresholds in volts, filter-settling-time vs. control-cycle timing, and ADC normalization assumptions that the RL environment relies on.	Full agreement. See §9.

3. Data-Integrity Fixes — Before Anything Else Is Written

These are not optional polish. Submitting with any of these unresolved is the kind of thing that gets a paper rejected on integrity grounds rather than merit grounds, which is a worse outcome than a novelty rejection.

Issue	Fix	Why it can't wait
“V4 Seed 3” reported alone as “the authoritative model”	Report the 5-seed aggregate (MAE 2205 ± 518 Hz, $84.8\% \pm 7.6\%$ near) as the primary result everywhere. Seed 3 may still be named as the best-observed seed / the checkpoint chosen for any future deployment, but never presented as if it were the typical result.	Selective reporting of the best-of-N run, without disclosing N or the spread, is the single fastest way to lose a reviewer's trust in every other number in the paper.
QACT v4/v5 numbers don't reproduce on rerun	Do not cite the currently saved QACT JSON numbers. Rerun the (fixed, see §8) pipeline with seeds fixed and report only numbers you can regenerate on demand.	An unreproducible number in a methods-focused case-study paper undermines the entire premise of the paper.
phase1_baseline.py / sweep_verify_v2.py import a nonexistent qiskit_resonator_env_v2	Either restore/recreate that module or repoint the scripts at the current env version, and confirm both run clean.	Dead code paths in a repository you may be asked to share alongside the paper (reproducibility is increasingly expected) are an easy, avoidable red flag.
Hardcoded C:\MiniProject_Sem4\... paths (actual project is at a different path)	Replace with relative paths from the repo root.	Same as above — trivial to fix, embarrassing to leave.
PID gain search and QACT runs are unseeded (results differ run to run)	Fix the RNG seed for the PID grid search, the classical env, and the AerSimulator; report mean $\pm$ std over ≥ 5 seeds for every result that goes in the paper, not just V4.	A results section where numbers change on rerun is not publishable regardless of framing.
pid_baseline.py and compare_v3_v4.py hardcode results rather than computing them	Rewrite so the reported numbers are computed from saved logs/checkpoints at run time, not pasted in as literals.	A reviewer or examiner who runs your own comparison script and gets a script that just prints fixed numbers will reasonably ask what else is hardcoded.

4. The Reframed Thesis

Given everything above, here is a thesis statement that is honest, uses material you actually have, and does not depend on anything that isn't reproducible:

“Reinforcement-learning-based calibration has already been demonstrated on real superconducting qubit hardware (Sivak et al., 2022; Baum et al., 2021; Ding et al., 2023). This work does not attempt to repeat that demonstration. Instead, working entirely in simulation on a classical LC-resonator analogue of the resonance-tracking problem central to qubit calibration, we ask a narrower, still-useful question: which specific encodings of derivative/rate information in an RL agent's observation space are necessary and sufficient for reliable frequency tracking under different drift statistics, and how fragile is a trained policy to superficially-compatible but semantically-altered observation spaces when transferred across simulators? Using a controlled, single-variable, multi-seed ablation protocol against a PID baseline, we show that (1) the tracking benefit from a derivative-based observation feature is concentrated in high-drift, non-mean-reverting (Gaussian) conditions and is statistically indistinguishable from baseline under mean-reverting (Ornstein–Uhlenbeck) drift; (2) naively encoded derivative information (raw or sign-only) degrades performance below a no-derivative baseline, while a specific frequency-normalized encoding does not — a feature-engineering sensitivity result, not a generic ‘more information helps’ claim; and (3) transferring a classically-trained policy into an independently implemented Qiskit-based quantum-circuit simulation fails primarily

due to silent observation-space semantic misalignment rather than any property of the physical target system — a general, reproducible pitfall for sim-to-sim and sim-to-real policy transfer wherever observation-space dimensionality is checked but semantics is not.”

Why this is stronger than either the original “novel discovery” framing or my first, more generic ‘case study’ draft: every one of its three claims is (a) something you can currently reproduce or will be able to once §3 and §6–§8 are done, (b) specific enough that a reviewer can check it rather than argue with a vague framing, and (c) positioned explicitly against real, named prior work rather than left implicit.

5. Updated Related-Work Table

Merged from both reviews — use this as the backbone of the related-work section:

Work	Venue / Year	Why it belongs in your related work
Fösel, Tighineanu, Weiss, Marquardt	Phys. Rev. X 8, 031084 (2018)	Foundational RL-for-quantum-feedback paper — establishes the field is ~8 years old.
Sivak et al., “Model-free quantum control with reinforcement learning”	Phys. Rev. X 12, 011059 (2022)	Your closest real-hardware comparator — cite explicitly in the thesis/limitations to pre-empt the “why not real qubits” question.
Baum et al.	PRX Quantum 2, 040324 (2021)	RL trained and validated on real superconducting hardware for gate-set design.
Ding et al., two-qubit fluxonium gate paper	arXiv:2304.06087 (2023)	PPO specifically, on real hardware, for a calibration task — directly pre-empts “why PPO” and “why not real hardware.”
Porotti, Essig, Royer, Poilblanc, Marquardt, “Deep RL for quantum state prep with weak nonlinear measurements”	Quantum 3, 173 (2019)	Your closest simulation-only comparator. Say explicitly where you exceed it (ablation depth, multi-seed reporting, classical baseline, dual-noise-regime testing, formal statistical testing) — this is a real, specific, checkable positioning claim.
Tobin et al., “Domain randomization for transferring deep neural networks from simulation to the real world”	IROS (2017)	Frames why observation/physics mismatches break transfer — directly supports Finding 3 of your thesis.
Zhao, Queralta, Westerlund, “Sim-to-Real Transfer in Deep RL for Robotics: A Survey”	IEEE SSCI (2020)	Same purpose — positions your QACT finding within an established robustness literature rather than as an isolated anomaly.
Ng, Harada, Russell, “Policy invariance under reward transformations”	ICML (1999)	Theoretical basis for the reward-vs-observation confound discussed in §6 — cite when presenting that ablation.

6. Missing Experiment: Separating Reward-Shaping From Observation-Augmentation

This is the single highest-priority experiment that doesn't exist yet, and it's cheap (~30–50 minutes per run on your RTX 3050). The problem: V4's reward already contains a gradient-shaped term

$(0.5 \cdot (ema_amp - prev_ema_amp) \cdot sign(action))$, so “adding the gradient to the observation” is not a clean, isolated intervention — you may be observing the effect of redundant/reinforcing signal, not new information. Run:

- V4's observation (with `amp_gradient`) but V3's reward (gradient bonus removed) — isolates whether the observation feature helps on its own.
- V3's observation (no gradient) but V4's reward (gradient bonus present) — isolates whether the reward shaping alone accounts for the V3→V4 gap.

Run each across the same 5 seeds and the same matched-1000-episode Mann-Whitney protocol already used for V3 vs. V4. Whatever the result, report it — a paper that shows “we checked whether this was really about the observation and not the reward, and here's what we found” is a stronger, more citable methodology contribution than the uncontrolled V3-vs-V4 comparison alone, independent of which way the result falls.

7. Leading With the Negative-Information Ablation (V4a / V4b)

Both reviews independently flagged this as the most underused result in the project, and I agree it should be the paper's headline finding rather than a footnote:

Variant	Gradient encoding	MAE (Hz)	Near-resonance	Low-amplitude
V3 (no gradient)	— (baseline)	3333 ± 1932	64.5%	13.4%
V4a	Raw Δamp, unnormalized	4007 ± 2311	55.2%	18.4%
V4b	sign(Δamp) only, 1 bit	3793 ± 2051	55.4%	17.8%
V4 (5-seed aggregate)	Δamp·3000/(Δfreq+1), normalized	2205 ± 518	84.8% ± 7.6%	5.1% ± 3.7%

This table, run across matched seeds with confidence intervals rather than the single-run numbers shown here, is your real contribution: **adding derivative information to the observation does not monotonically help — the naive encodings are actively worse than having no derivative term at all, and only a specific frequency-normalized form recovers and exceeds baseline.** That's a feature-engineering-sensitivity finding with a plausible mechanism (raw Δamp has the wrong scale relative to the policy's other inputs; sign-only discards magnitude entirely), it's specific enough to be checked, and it doesn't depend on the QACT pipeline being fixed at all — you already have the checkpoints.

8. Fixing and Rerunning the Qiskit / QACT Study

Consolidated, priority-ordered, from both reviews—with the isolation-experiment evidence folded in:

Step	Action	Evidence it will help
1	Align the observation vector exactly to V4's order and scaling: [norm_freq, ema_amp, prev_ema_amp, prev_action, amp_gradient], gradient clipped to ± 1 , computed as $\Delta \text{amp} \cdot 3000 / (\Delta \text{freq} + 1)$.	Proven: this single fix took zero-shot MAE from 2964 ± 2473 to 1708 ± 1103 Hz and near-resonance from 32.8% to 66.5%, with no retraining.
2	Match the resonance linewidth: raise omega_rabi to ≈ 5000 Hz so the 0.90 crossing (± 3813 Hz) matches the classical ± 3333 Hz band — or keep 2500 Hz but report a width-normalized metric and disclose the asymmetry explicitly.	Measured: at omega_rabi=2500, the Qiskit env's near-resonance metric is $\sim 1.75\times$ stricter than the classical one for nominally the same threshold.
3	Port the full shaped reward (gradient bonus, action-cost penalty, low-amp penalty, 3-step dead termination) instead of reward = amp alone.	The value function was trained against this reward; handing it an unshaped one changes the MDP the policy is being evaluated against.
4	Restore action_scale = 3000 (currently cut to 1000 in the Qiskit env).	At 1000, the agent is under-agile relative to a 500 Hz/step drift rate on a narrow curve — a scale mismatch, not a transfer failure.
5	Fix reproducibility: restore/repoint qiskit_resonator_env_v2, replace hardcoded paths, seed the AerSimulator and every RNG.	Currently the saved v4/v5 numbers don't reproduce on rerun — nothing downstream is trustworthy until this is fixed.
6	Retrain/fine-tune properly: multiple parallel envs (not one DummyVecEnv), $\geq 500k$ steps (not 125–150k), ≥ 5 seeds, 4096+ shots per evaluation.	The current v4-vs-v5 spread (2176 vs. 3237 Hz from “nearly identical” runs, stds larger than the means) is consistent with an under-budgeted, unstable fine-tune, not a stable estimate of transfer difficulty.
7	Report zero-shot and fine-tuned numbers side by side, both with the fully aligned setup.	This is the actual scientific question — “does the policy transfer across amplitude models when everything else is held constant” — rather than a confounded comparison.

With steps 1–6 done, a reasonable, testable expectation — not a promise — is a zero-shot MAE in the neighborhood of 1500–1800 Hz with near-resonance around 65–75%, based directly on the isolation test already run. That would be a genuinely strong, honestly-earned result for the paper's transfer section.

9. Multisim / Simulink — Scope Confirmed

Both reviews agree, and the agentic audit sharpens it into concrete deliverables. Use Multisim/Simulink to validate the assumptions the RL environment is built on, not to substitute for the RL results themselves:

- Confirm the Lorentzian amplitude-response shape ($Q=50$, $f_0=500$ kHz) against an independent circuit simulator, not just LTSpice.
- Confirm the 0.90 / 0.70 amplitude thresholds used in the RL reward/metrics correspond to sensible values in real volts at the envelope detector output.
- Confirm the RC low-pass filter's settling time is compatible with the ~ 50 ms control cycle the RL agent assumes — if settling is slower than the control cycle, the simulated amplitude signal is unrealistically clean/fast.
- Confirm the ADC normalization assumptions (MCP3202, 12-bit) match what the simulation treats as the amplitude observation's dynamic range.

Frame this precisely in the paper as: “the analog front-end's assumptions (resonance shape, thresholds, filter timing, ADC scaling) were verified against an independent circuit simulator; the RL policy itself was evaluated only in simulation.” That is honest, adds real credibility, and does not overreach into a hardware-validation claim you can't support.

10. Updated Claims: Avoid vs. Use

Avoid	Use instead
“V4 Seed 3 achieves MAE 1510 Hz” as the headline result	“V4 achieves MAE 2205 ± 518 Hz (5-seed aggregate); the best-observed seed reached 1510 ± 367 Hz”
“Adding the amplitude gradient to the observation improves tracking” (unqualified)	“A frequency-normalized gradient feature improves tracking; naive (raw or sign-only) encodings of the same underlying signal degrade it below baseline” — and, once §6 is run, a statement about whether this is separable from the reward's own gradient term
“Sim-to-real transfer” for the QACT experiment	“Cross-simulator transfer” or “sim-to-sim transfer,” evaluated against an independently implemented Qiskit-based simulation
Reporting the currently saved QACT v4/v5 numbers	Reporting only numbers from a rerun of the aligned, seeded pipeline (§8) that you can regenerate on request
“Robust across noise regimes”	“Benefit is concentrated in high-drift, non-mean-reverting (Gaussian) conditions; statistically indistinguishable from baseline under mean-reverting (OU) drift”
Any remaining hardware Phase 1–5 roadmap language as near-term	A single, explicit limitations-section statement: physical hardware validation was out of scope; all results are simulation-based, with the analog front-end cross-validated in §9

11. Paper Structure for the Case Study

Section	Contents
1. Introduction	Motivate via real-hardware RL calibration literature (Sivak, Baum, Ding); state plainly that this work does not attempt to replicate that on real qubits, and state the narrower thesis from §4.
2. Related Work	The table in §5, organized as: real-hardware RL-for-qubit-calibration (context/motivation, not a target to beat) vs. simulation-only comparators (Porotti — a target you can concretely compare against) vs. sim-to-real/sim-to-sim transfer literature (frames §6's finding).
3. Classical Testbed and Methodology	Resonator model, PID baseline, V3/V4/V4a/V4b architectures, training protocol, Mann-Whitney matched-episode evaluation protocol — written so it's independently reproducible.
4. Results I — Gradient Feature Ablation	The V3/V4a/V4b/V4 table from §7 as the headline result, the §6 confound-clearing ablation, PID comparison (mean-and-variance argument), Gaussian-vs-OU regime split with the honest null result.
5. Results II — Cross-Simulator Transfer	The QACT setup, the isolation-experiment methodology, the misaligned-vs-aligned comparison table, and the zero-shot/fine-tuned numbers from the properly rerun pipeline (§8).
6. Discussion	What generalizes: the feature-engineering-sensitivity finding (§7) and the semantic-misalignment pitfall (§6/§8) as lessons for RL-based control/transfer work generally, not just this testbed.
7. Limitations and Future Work	Hardware out of scope (state once, plainly); Multisim/Simulink front-end validation (§9) as what was done instead; note the toy-environment scope honestly.

12. Final Action Plan and Phased Timeline

Phase	What	Depends on	Est. effort
0 — Integrity fixes	All six items in §3: report the aggregate not Seed 3 alone, stop citing unreproducible QACT numbers, fix dead imports/paths, seed everything, de-hardcode the comparison scripts.	Nothing — start here	1–2 days
1 — Confound ablation	Run the two §6 experiments (V4 obs + V3 reward; V3 obs + V4 reward) across 5 seeds each with the existing Mann-Whitney protocol.	Phase 0	~1 day compute + analysis
2 — QACT rebuild	Apply all seven §8 fixes; retrain/fine-tune with adequate budget and ≥ 5 seeds; report zero-shot and fine-tuned numbers.	Phase 0	2–4 days, mostly compute
3 — Ablation table finalization	Re-run V4a/V4b/V4/V3 across matched seeds with confidence intervals for the §7 headline table.	Phase 0	~1 day compute
4 — Front-end validation (optional but recommended)	Multisim/Simulink checks from §9.	Can run in parallel with Phases 1–3	1–2 days
5 — Writing	Draft using the §11 structure, the §4 thesis statement, and the §10 claims table as a running checklist while writing.	Phases 0–3 complete (Phase 4 improves but doesn't block it)	Depends on pace — sequence last

13. Updated Publishability Estimate

My original range (roughly 30–55% at a venue like IEEE Access, contingent on fixes) still holds as a shape, but the basis for it is now considerably firmer. The case-study reframing removes the novelty risk almost entirely — you're no longer claiming a discovery a competent reviewer could look up and find already done. What now determines the outcome is narrower and more within your control: whether §3's integrity fixes are actually made, whether §6's confound ablation is run and reported honestly whichever way it comes out, and whether §8's QACT rerun produces numbers you can stand behind on request. None of that is guaranteed to go your way scientifically — the confound ablation, in particular, might show the observation feature adds little beyond what the reward already provides, and if so, that has to be reported as such, not around. But a paper that does all of this and reports it straight, even with a mixed result in §6, is a materially more publishable document than either the original draft or my first-pass reframing, precisely because “rigorous case study” is a bar this version can actually clear on the merits rather than on careful wording.